Question ID:50111
Topic Name: Mathematics-Section A

Question: Let α be a root of the equation $1 + x^2 + x^4 = 0$. Then the value of $\alpha^{1011} + \alpha^{2022} - \alpha^{3033}$ is equal to :

A	1
B	α
C	$1 + \alpha$
D	$1 + 2\alpha$

Answer Given By Candidate:**A**

Question ID:50112
Topic Name: Mathematics-Section A

Question: Let $\arg(z)$ represent the principal argument of the complex number z. Then, $|z| = 3$ and $\arg(z - 1) - \arg(z + 1) = \frac{\pi}{4}$ intersect

A	exactly at one point.
B	exactly at two points.
C	nowhere.
D	at infinitely many points.

Answer Given By Candidate:**Not Attempted**

Question ID:50113
Topic Name: Mathematics-Section A

Let $A=\begin{pmatrix}2 & -1\\ 0 & 2\end{pmatrix}$. If $B=I-{}^5C_1\,(adjA)+{}^5C_2\,(adjA)^2-\ldots-{}^5C_5\,(adjA)^5$, then the

Question: sum of all elements of the matrix B is

A	-5
B	-6
C	-7
D	-8

Answer Given By Candidate:**Not Attempted**

Question ID:50114
Topic Name: Mathematics-Section A

The sum of the infinite series $1+\frac{5}{6}+\frac{12}{6^2}+\frac{22}{6^3}+\frac{35}{6^4}+\frac{51}{6^5}+\frac{70}{6^6}+\ldots$

Question: is equal to:

A	$\frac{425}{216}$
B	$\frac{429}{216}$
C	$\frac{288}{125}$
D	$\frac{280}{125}$

Answer Given By Candidate:**C**

Question ID:50115
Topic Name: Mathematics-Section A

The value of $\lim\limits_{x\to 1}\frac{\left(x^2-1\right)\sin^2(\pi x)}{x^4-2x^3+2x-1}$

Question: is equal to:

A	$\frac{\pi^2}{6}$
B	

	$\frac{\pi^2}{3}$
C	$\frac{\pi^2}{2}$
D	π^2

Answer Given By Candidate:**Not Attempted**

Question ID:50116
Topic Name: Mathematics-Section A

Let $f : \mathbb{R} \to \mathbb{R}$ be a function defined by $f(x) = (x-3)^{n_1}(x-5)^{n_2}$, $n_1, n_2 \in \mathbb{N}$.

Question: Then, which of the following is NOT true?

A	For $n_1 = 3, n_2 = 4$, there exists $\alpha \in (3,5)$ where f attains local maxima.
B	For $n_1 = 4, n_2 = 3$, there exists $\alpha \in (3,5)$ where f attains local minima.
C	For $n_1 = 3, n_2 = 5$, there exists $\alpha \in (3,5)$ where f attains local maxima.
D	For $n_1 = 4, n_2 = 6$, there exists $\alpha \in (3,5)$ where f attains local maxima.

Answer Given By Candidate:**Not Attempted**

Question ID:50117
Topic Name: Mathematics-Section A

Let f be a real valued continuous function on $[0, 1]$ and

$$f(x) = x + \int_0^1 (x-t)f(t)dt.$$

Question: Then, which of the following points (x, y) lies on the curve $y = f(x)$?

A	$(2, 4)$
B	$(1, 2)$
C	$(4, 17)$
D	$(6, 8)$

Answer Given By Candidate:**Not Attempted**

Question ID:50118
Topic Name: Mathematics-Section A

If $\int_0^2\left(\sqrt{2x}-\sqrt{2x-x^2}\right)dx=\int_0^1\left(1-\sqrt{1-y^2}-\frac{y^2}{2}\right)dy+\int_1^2\left(2-\frac{y^2}{2}\right)dy+I$

Question: then I equals

A $\int_0^1\left(1+\sqrt{1-y^2}\right)dy$

B $\int_0^1\left(\frac{y^2}{2}-\sqrt{1-y^2}+1\right)dy$

C $\int_0^1\left(1-\sqrt{1-y^2}\right)dy$

D $\int_0^1\left(\frac{y^2}{2}+\sqrt{1-y^2}+1\right)dy$

Answer Given By Candidate:**B**

Question ID:50119

Topic Name: Mathematics-Section A

If $y=y(x)$ is the solution of the differential equation $\left(1+e^{2x}\right)\frac{dy}{dx}+2\left(1+y^2\right)e^x=0$ and $y(0)=0$, then $6\left(y'(0)+\left(y\left(\log_e\sqrt{3}\right)\right)^2\right)$ is equal to

Question:

A 2

B -2

C -4

D -1

Answer Given By Candidate:**B**

Question ID:501110

Topic Name: Mathematics-Section A

Let P : $y^2 = 4ax$, $a > 0$ be a parabola with focus S. Let the tangents to the parabola P make an angle of $\frac{\pi}{4}$ with the line $y = 3x + 5$ touch the parabola P at A and B.

Question: Then the value of a for which A, B and S are collinear is

A 8 only

B 2 only

C $\frac{1}{4}$ only

D any $a > 0$

Answer Given By Candidate:**Not Attempted**

Question ID:501111

Topic Name: Mathematics-Section A

Question: Let a triangle ABC be inscribed in the circle $x^2 - \sqrt{2}(x+y) + y^2 = 0$ such that $\angle BAC = \frac{\pi}{2}$. If the length of side AB is $\sqrt{2}$, then the area of the ΔABC is equal to :

A $\left(\sqrt{2}+\sqrt{6}\right)/3$

B $\left(\sqrt{6}+\sqrt{3}\right)/2$

C $\left(3+\sqrt{3}\right)/4$

D $\left(\sqrt{6}+2\sqrt{3}\right)/4$

Answer Given By Candidate:**D**

Question ID:501112

Topic Name: Mathematics-Section A

Question: Let $\frac{x-2}{3} = \frac{y+1}{-2} = \frac{z+3}{-1}$ lie on the plane $px - qy + z = 5$, for some $p, q \in \mathbb{R}$. The shortest distance of the plane from the origin is:

A $\sqrt{\frac{3}{109}}$

B $\sqrt{\frac{5}{142}}$

C $\frac{5}{\sqrt{71}}$

D	$\frac{1}{\sqrt{142}}$

Answer Given By Candidate:**C**

Question ID:501113
Topic Name: Mathematics-Section A

Question:

The distance of the origin from the centroid of the triangle whose two sides have the equations

$x-2y+1=0$ and $2x-y-1=0$ and whose orthocenter is $\left(\frac{7}{3},\frac{7}{3}\right)$ is:

A	$\sqrt{2}$
B	2
C	$2\sqrt{2}$
D	4

Answer Given By Candidate:**Not Attempted**

Question ID:501114
Topic Name: Mathematics-Section A

Question:

Let Q be the mirror image of the point P(1, 2, 1) with respect to the plane $x+2y+2z=16$. Let T be a plane passing through the point Q and contains the line $\vec{r}=-\hat{k}+\lambda\left(\hat{i}+\hat{j}+2\hat{k}\right), \lambda\in\mathbb{R}$. Then, which of the following points lies on T?

A	(2, 1, 0)
B	(1, 2, 1)
C	(1, 2, 2)
D	(1, 3, 2)

Answer Given By Candidate:**C**

Question ID:501115
Topic Name: Mathematics-Section A

Question: Let A, B, C be three points whose position vectors respectively are

$\vec{a} = \hat{i} + 4\hat{j} + 3\hat{k}$

$\vec{b} = 2\hat{i} + \alpha\hat{j} + 4\hat{k}, \alpha \in \mathbb{R}$

$\vec{c} = 3\hat{i} - 2\hat{j} + 5\hat{k}$

If α is the smallest positive integer for which $\vec{a}$, $\vec{b}$, $\vec{c}$ are noncollinear, then the length of the median, in ΔABC, through A is :

A $\frac{\sqrt{82}}{2}$

B $\frac{\sqrt{62}}{2}$

C $\frac{\sqrt{69}}{2}$

D $\frac{\sqrt{66}}{2}$

Answer Given By Candidate:**C**

Question ID:501116
Topic Name: Mathematics-Section A

Question: The probability that a relation R from $\{x, y\}$ to $\{x, y\}$ is both symmetric and transitive, is equal to

A $\frac{5}{16}$

B $\frac{9}{16}$

C $\frac{11}{16}$

D $\frac{13}{16}$

Answer Given By Candidate:**A**

Question ID:501117
Topic Name: Mathematics-Section A

Question: The number of values of $a \in N$ such that the variance of $3, 7, 12, a, 43 - a$ is a natural number is:

A	0
B	2
C	5
D	infinite

Answer Given By Candidate:**Not Attempted**

Question ID:501118
Topic Name: Mathematics-Section A
Question: From the base of a pole of height 20 meter, the angle of elevation of the top of a tower is 60°. The pole subtends an angle 30° at the top of the tower. Then the height of the tower is:

A	$15\sqrt{3}$
B	$20\sqrt{3}$
C	$20 + 10\sqrt{3}$
D	30

Answer Given By Candidate:**Not Attempted**

Question ID:501119
Topic Name: Mathematics-Section A
Question: Negation of the Boolean statement $(p \vee q) \Rightarrow ((\sim r) \vee p)$ is equivalent to

A	$p \wedge (\sim q) \wedge r$
B	$(\sim p) \wedge (\sim q) \wedge r$
C	$(\sim p) \wedge q \wedge r$
D	$p \wedge q \wedge (\sim r)$

Answer Given By Candidate:**Not Attempted**

Question ID:501120
Topic Name: Mathematics-Section A
Question: Let $n \geq 5$ be an integer. If $9^n - 8n - 1 = 64\alpha$ and $6^n - 5n - 1 = 25\beta$, then $\alpha - \beta$ is equal to

A $1+{}^{n}C_{2}(8-5)+{}^{n}C_{3}(8^{2}-5^{2})+\ldots+{}^{n}C_{n}(8^{n-1}-5^{n-1})$

B $1+{}^{n}C_{3}(8-5)+{}^{n}C_{4}(8^{2}-5^{2})+\ldots+{}^{n}C_{n}(8^{n-2}-5^{n-2})$

C ${}^{n}C_{3}(8-5)+{}^{n}C_{4}(8^{2}-5^{2})+\ldots+{}^{n}C_{n}(8^{n-2}-5^{n-2})$

D ${}^{n}C_{4}(8-5)+{}^{n}C_{5}(8^{2}-5^{2})+\ldots+{}^{n}C_{n}(8^{n-3}-5^{n-3})$

Answer Given By Candidate:**A**

Question ID:501121
Topic Name: Mathematics-Section B
Question:

Let $\vec{a}=\hat{i}-2\hat{j}+3\hat{k}$, $\vec{b}=\hat{i}+\hat{j}+\hat{k}$ and $\vec{c}$ be a vector such that $\vec{a}+(\vec{b}\times\vec{c})=\vec{0}$ and $\vec{b}\cdot\vec{c}=5$. Then, the value of $3(\vec{c}\cdot\vec{a})$ is equal to ___.

Answer Given By Candidate:**18**

Question ID:501122
Topic Name: Mathematics-Section B

Let $y=y(x)$, $x>1$, be the solution of the differential equation $(x-1)\frac{dy}{dx}+2xy=\frac{1}{x-1}$, with $y(2)=\frac{1+e^{4}}{2e^{4}}$. If $y(3)=\frac{e^{\alpha}+1}{\beta e^{\alpha}}$, then the value of

Question: $\alpha+\beta$ is equal to __.

Answer Given By Candidate:**Not Attempted**

Question ID:501123
Topic Name: Mathematics-Section B

Let 3, 6, 9, 12,… upto 78 terms and 5, 9, 13, 17,.. upto 59 terms be two series.

Question: Then, the sum of the terms common to both the series is equal to__.

Answer Given By Candidate:**2223**

Question ID:501124
Topic Name: Mathematics-Section B

Question: The number of solutions of the equation $\sin x=\cos^{2}x$ in the interval (0, 10) is __.

Answer Given By Candidate:**Not Attempted**

Question ID:501125
Topic Name: Mathematics-Section B

Question: For real numbers a, b $(a > b > 0)$, let

$$\text{Area}\left\{(x, y): x^2 + y^2 \le a^2 \text{ and } \frac{x^2}{a^2} + \frac{y^2}{b^2} \ge 1\right\} = 30\pi$$

and

$$\text{Area}\left\{(x, y): x^2 + y^2 \ge b^2 \text{ and } \frac{x^2}{a^2} + \frac{y^2}{b^2} \le 1\right\} = 18\pi$$

Then the value of $(a - b)^2$ is equal to __.

Answer Given By Candidate:**Not Attempted**

Question ID:501126
Topic Name: Mathematics-Section B

Question: Let f and g be twice differentiable even functions on $(-2, 2)$ such that

$$f\left(\frac{1}{4}\right) = 0,\ f\left(\frac{1}{2}\right) = 0,\ f(1) = 1 \text{ and } g\left(\frac{3}{4}\right) = 0,\ g(1) = 2$$

Then, the minimum number of solutions of $f(x)g''(x) + f'(x)g'(x) = 0$ in $(-2, 2)$ is equal to___.

Answer Given By Candidate:**Not Attempted**

Question ID:501127
Topic Name: Mathematics-Section B

Question: Let the coefficients of x^{-1} and x^{-3} in the expansion of $\left(2x^{\frac{1}{5}} - \frac{1}{x^{\frac{1}{5}}}\right)^{15}$, $x > 0$, be m and n respectively. If r is a positive integer such that $mn^2 = {}^{15}C_r \cdot 2^r$, then the value of r is equal to ___.

Answer Given By Candidate:**Not Attempted**

Question ID:501128
Topic Name: Mathematics-Section B

Question: The total number of four digit numbers such that each of first three digits is divisible by the last digit, is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:501129
Topic Name: Mathematics-Section B

Question: Let $M = \begin{bmatrix} 0 & -\alpha \\ \alpha & 0 \end{bmatrix}$, where α is a non-zero real number an $N = \sum_{k=1}^{49} M^{2k}$. If $(I - M^2)N = -2I$, then the positive integral value of α is ___.

Answer Given By Candidate:**Not Attempted**

Question ID:501130
Topic Name: Mathematics-Section B

Question: Let $f(x)$ and $g(x)$ be two real polynomials of degree 2 and 1 respectively. If $f(g(x)) = 8x^2 - 2x$, and $g(f(x)) = 4x^2 + 6x + 1$, then the value of $f(2) + g(2)$ is _____.

Answer Given By Candidate:**Not Attempted**

Question ID:501131
Topic Name: Physics-Section A
Question:
A small toy starts moving from the position of rest under a constant acceleration. If it travels a distance of 10m in t s, the distance travelled by the toy in the next t s will be :

A 10m

B 20m

C 30m

D 40m

Answer Given By Candidate:**B**

Question ID:501132
Topic Name: Physics-Section A

Question: At what temperature a gold ring of diameter 6.230 cm be heated so that it can be fitted on a wooden bangle of diameter 6.241 cm ? Both the diameters have been measured at room temperature (27°C).
(Given: coefficient of linear thermal expansion of gold $\alpha_L = 1.4 \times 10^{-5}\ K^{-1}$)

A 125.7°C

B 91.7°C

C 425.7°C

D 152.7°C

Answer Given By Candidate:**A**

Question ID:501133
Topic Name: Physics-Section A
Question: Two point charges Q each are placed at a distance d apart. A third point charge q is placed at a distance x from mid-point on the perpendicular bisector. The value of x at which charge q will experience the maximum Coulomb's force is:

A $x = d$

B $x = \frac{d}{2}$

C $x = \frac{d}{\sqrt{2}}$

D $x = \frac{d}{2\sqrt{2}}$

Answer Given By Candidate:**B**

Question ID:501134
Topic Name: Physics-Section A
Question:
The speed of light in media 'A' and 'B' are 2.0×10^{10} cm/s and 1.5×10^{10} cm/s respectively. A ray of light enters from the medium B to A at an incident angle 'θ'. If the ray suffers total internal reflection, then

A $\theta = \sin^{-1}\left(\frac{3}{4}\right)$

B $\theta > \sin^{-1}\left(\frac{2}{3}\right)$

C $\theta < \sin^{-1}\left(\frac{3}{4}\right)$

D $\theta > \sin^{-1}\left(\frac{3}{4}\right)$

Answer Given By Candidate:**D**

Question ID:501135
Topic Name: Physics-Section A

Question: In the following nuclear reaction,

$D \xrightarrow{\alpha} D_1 \xrightarrow{\beta^-} D_2 \xrightarrow{\alpha} D_3 \xrightarrow{\gamma} D_4$

Mass number of D is 182 and atomic number is 74. Mass number and atomic number of D_4 respectively will be ___

A	174 and 71
B	174 and 69
C	172 and 69
D	172 and 71

Answer Given By Candidate:**A**

Question ID:501136

Topic Name: Physics-Section A

Question: The electric field at a point associated with a light wave is given by

$E = 200\ [\ \sin(6 \times 10^{15})\ t\ +\ \sin(9 \times 10^{15})\ t\ ]\text{Vm}^{-1}$

Given : $h = 4.14 \times 10^{-15}$ eVs

If this light falls on a metal surface having a work function of 2.50 eV, the maximum kinetic energy of the photoelectrons will be

A	1.90 eV
B	3.27 eV
C	3.60 eV
D	3.42 eV

Answer Given By Candidate:**Not Attempted**

Question ID:501137

Topic Name: Physics-Section A

Question: A capacitor is discharging through a resistor R. Consider in time t_1, the energy stored in the capacitor reduces to half of its initial value and in time t_2, the charge stored reduces to one eighth of its initial value. The ratio t_1/t_2 will be

A	1/2
B	1/3
C	1/4
D	1/6

Answer Given By Candidate:C

Question ID:501138
Topic Name: Physics-Section A
Question:
Starting with the same initial conditions, an ideal gas expands from volume V_1 to V_2 in three different ways. The work done by the gas is W_1 if the process is purely isothermal, W_2, if the process is purely adiabatic and W_3 if the process is purely isobaric. Then, choose the correct option

A	$W_1 < W_2 < W_3$
B	$W_2 < W_3 < W_1$
C	$W_3 < W_1 < W_2$
D	$W_2 < W_1 < W_3$

Answer Given By Candidate:D

Question ID:501139
Topic Name: Physics-Section A
Question:
Two long current carrying conductors are placed parallel to each other at a distance of 8 cm between them. The magnitude of magnetic field produced at mid-point between the two conductors due to current flowing in them is 300 μT . The equal current flowing in the two conductors is :

A	30A in the same direction.
B	30A in the opposite direction.
C	60A in the opposite direction.
D	300A in the opposite direction.

Answer Given By Candidate:B

Question ID:501140
Topic Name: Physics-Section A
Question:
The time period of a satellite revolving around earth in a given orbit is 7 hours. If the radius of orbit is increased to three times its previous value, then approximate new time period of the satellite will be

A	40 hours
B	36 hours

C	30 hours
D	25 hours

Answer Given By Candidate:**B**

Question ID:501141
Topic Name: Physics-Section A

Question: The TV transmission tower at a particular station has a height of 125 m. For doubling the coverage of its range, the height of the tower should be increased by

A	125 m
B	250 m
C	375 m
D	500 m

Answer Given By Candidate:**C**

Question ID:501142
Topic Name: Physics-Section A

Question: The motion of a simple pendulum executing S.H.M. is represented by the following equation.

$y = A\sin(\pi t + \phi)$, where time is measured in second.

The length of pendulum is

A	97.23 cm
B	25.3 cm
C	99.4 cm
D	406.1 cm

Answer Given By Candidate:**C**

Question ID:501143
Topic Name: Physics-Section A

Question: A vessel contains 16g of hydrogen and 128 g of oxygen at standard temperature and pressure. The volume of the vessel in cm^3 is :

A	72×10^5
B	32×10^5
C	27×10^4

D	54×10^4

Answer Given By Candidate:**B**

Question ID:501144
Topic Name: Physics-Section A

Question:

Given below are two statements :

Statement I: The electric force changes the speed of the charged particle and hence changes its kinetic energy; whereas the magnetic force does not change the kinetic energy of the charged particle.

Statement II: The electric force accelerates the positively charged particle perpendicular to the direction of electric field. The magnetic force accelerates the moving charged particle along the direction of magnetic field.

In the light of the above statements, choose the *most appropriate* answer from the options given below :

A	Both Statement I and Statement II are correct.
B	Both Statement I and Statement II are incorrect.
C	Statement I is correct but Statement II is incorrect.
D	Statement I is incorrect but Statement II is correct.

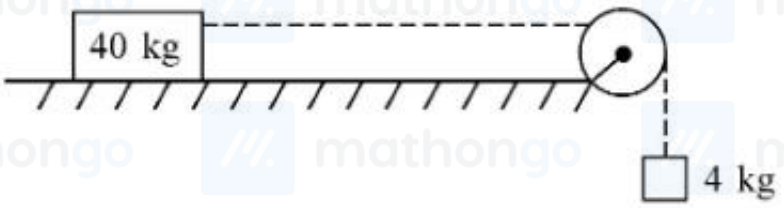
Answer Given By Candidate:**C**

Question ID:501145
Topic Name: Physics-Section A

Question:

A block of mass 40 kg slides over a surface, when a mass of 4 kg is suspended through an inextensible massless string passing over frictionless pulley as shown below.

The coefficient of kinetic friction between the surface and block is 0.02. The acceleration of block is. (Given g = 10 ms^{-2}.)



A	1 ms^{-2}
B	1/5 ms^{-2}
C	4/5 ms^{-2}

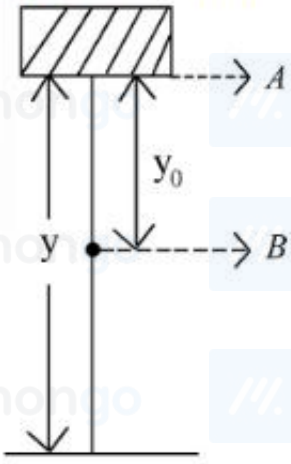
D $8/11\ ms^{-2}$

Answer Given By Candidate:**D**

Question ID:501146

Topic Name: Physics-Section A

In the given figure, the block of mass m is dropped from the point 'A'. The expression for kinetic energy of block when it reaches point 'B' is



Question:

A $\frac{1}{2}mg\,y_0^2$

B $\frac{1}{2}mg\,y^2$

C $mg(y-y_0)$

D mgy_0

Answer Given By Candidate:**C**

Question ID:501147

Topic Name: Physics-Section A

A block of mass M placed inside a box descends vertically with acceleration 'a'. The block exerts a force equal to one-fourth of its weight on the floor of the box.

Question: The value of 'a' will be

A $\frac{g}{4}$

B $\frac{g}{2}$

C $\frac{3g}{4}$

D	g

Answer Given By Candidate:**C**

Question ID:501148
Topic Name: Physics-Section A
Question: If the electric potential at any point (x, y, z) m in space is given by $V = 3x^2$ volt. The electric field at the point $(1, 0, 3)$ m will be :

A	3 Vm^{-1}, directed along positive x-axis.
B	3 Vm^{-1}, directed along negative x-axis.
C	6 Vm^{-1}, directed along positive x-axis.
D	6 Vm^{-1}, directed along negative x-axis.

Answer Given By Candidate:**D**

Question ID:501149
Topic Name: Physics-Section A
Question: The combination of two identical cells, whether connected in series or parallel combination provides the same current through an external resistance of 2Ω. The value of internal resistance of each cell is

A	2Ω
B	4Ω
C	6Ω
D	8Ω

Answer Given By Candidate:**A**

Question ID:501150
Topic Name: Physics-Section A
Question: A person can throw a ball upto a maximum range of 100 m. How high above the ground he can throw the same ball?

A	25 m
B	50 m
C	100 m
D	200 m

Answer Given By Candidate:C

Question ID:501151
Topic Name: Physics-Section B

Question: The Vernier constant of Vernier callipers is 0.1 mm and it has zero error of (−0.05) cm. While measuring diameter of a sphere, the main scale reading is 1.7 cm and coinciding vernier division is 5. The corrected diameter will be _____ $\times 10^{-2}$ cm.

Answer Given By Candidate:**Not Attempted**

Question ID:501152
Topic Name: Physics-Section B

Question: A small spherical ball of radius 0.1 mm and density 10^4 kg m^{-3} falls freely under gravity through a distance h before entering a tank of water. If, after entering the water the velocity of ball does not change and it continue to fall with same constant velocity inside water, then the value of h will be _____ m (Given g = 10 ms^{-2}, viscosity of water = 1.0×10^{-5} N-sm^{-2}).

Answer Given By Candidate:**Not Attempted**

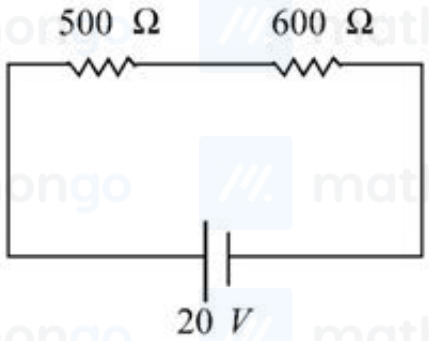
Question ID:501153
Topic Name: Physics-Section B

Question: In an experiment to determine the velocity of sound in air at room temperature using a resonance tube, the first resonance is observed when the air column has a length of 20.0 cm for a tuning fork of frequency 400 Hz is used. The velocity of the sound at room temperature is 336 ms^{-1}. The third resonance is observed when the air column has a length of _____cm

Answer Given By Candidate:**Not Attempted**

Question ID:501154
Topic Name: Physics-Section B

Question: Two resistors are connected in series across a battery as shown in figure. If a voltmeter of resistance 2000 Ω is used to measure the potential difference across 500 Ω resister, the reading of the voltmeter will be _____V



Answer Given By Candidate:**3**

Question ID:501155
Topic Name: Physics-Section B

Question: A potential barrier of 0.4 V exists across a p-n junction. An electron enters the junction from the n-side with a speed of $6.0 \times 10^5 ms^{-1}$. The speed with which electron enters the p side will be $\frac{x}{3} \times 10^5\ ms^{-1}$ the value of x is _____.
(Given mass of electron = 9×10^{-31} kg, charge on electron = 1.6×10^{-19} C.)

Answer Given By Candidate:**Not Attempted**

Question ID:501156
Topic Name: Physics-Section B

Question: The displacement current of 4.425 μA is developed in the space between the plates of parallel plate capacitor when voltage is changing at a rate of $10^6\ Vs^{-1}$. The area of each plate of the capacitor is 40 cm^2. The distance between each plate of the capacitor is $x \times 10^{-3}$m.The value of x is ,
(Permittivity of free space, $E_0 = 8.85 \times 10^{-12}\ C^2\ N^{-1}\ m^{-2}$) _____

Answer Given By Candidate:**Not Attempted**

Question ID:501157
Topic Name: Physics-Section B

Question: The moment of inertia of a uniform thin rod about a perpendicular axis passing through one end is I_1 . The same rod is bent into a ring and its moment of inertia about a diameter is I_2 . If $\frac{I_1}{I_2}$ is $\frac{x\pi^2}{3}$, then the value of x will be _____.

Answer Given By Candidate:**8**

Question ID:501158
Topic Name: Physics-Section B

Question: The half life of a radioactive substance is 5 years. After x years a given sample of the radioactive substance gets reduced to 6.25% of its initial value. The value of x is _____

Answer Given By Candidate:**Not Attempted**

Question ID:501159
Topic Name: Physics-Section B

Question: In a double slit experiment with monochromatic light, fringes are obtained on a screen placed at some distance from the plane of slits. If the screen is moved by 5×10^{-2} m towards the slits, the change in fringe width is 3×10^{-3} cm. If the distance between the slits is 1 mm, then the wavelength of the light will be _____ nm.

Answer Given By Candidate:**600**

Question ID:501160
Topic Name: Physics-Section B

Question: An inductor of 0.5 mH, a capacitor of 200 μF and a resistor of 2 Ω are connected in series with a 220 V ac source. If the current is in phase with the emf, the frequency of ac source will be _____ $\times 10^2$ Hz.

Answer Given By Candidate:**5**

Question ID:501161
Topic Name: Chemistry-Section A

Question: Using the rules for significant figures, the correct answer for the expression $\frac{0.02858 \times 0.112}{0.5702}$ will be

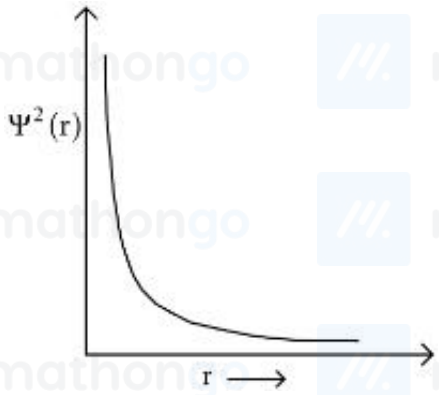
A	0.005613
B	0.00561
C	0.0056
D	0.006

Answer Given By Candidate:**C**

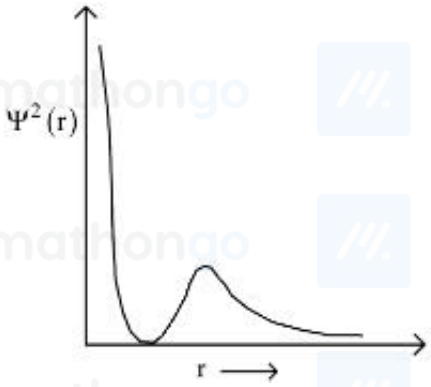
Question ID:501162
Topic Name: Chemistry-Section A

Question: Which of the following is the correct plot for the probability density $\Psi^2(r)$ as a function of distance 'r' of the electron from the nucleus for 2s orbital?

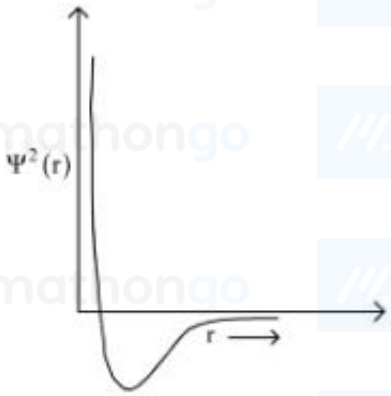
A



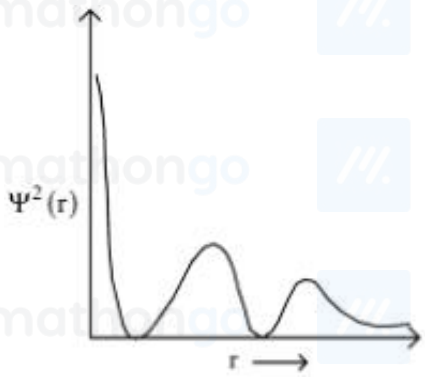
B



C



D



Answer Given By Candidate:**D**

Question ID:501163
Topic Name: Chemistry-Section A

Question:
Consider the species CH_4, NH_4^+ and BH_4^-. Choose the correct option with respect to the there species.

A	They are isoelectronic and only two have tetrahedral structures.
B	They are isoelectronic and all have tetrahedral structures.
C	Only two are isoelectronic and all have tetrahedral structures.
D	Only two are isoelectronic and only two have tetrahedral structures.

Answer Given By Candidate:**B**

Question ID:501164
Topic Name: Chemistry-Section A
Question:
4.0 moles of argon and 5.0 moles of PCl_5 are introduced into an evacuated flask of 100 litre capacity at 610 K. The system is allowed to equilibrate. At equilibrium, the total pressure of mixture was found to be 6.0 atm. The K_p for the reaction is
[Given: R = 0.082 L atm K^{-1} mol^{-1}]

A	2.25
B	6.24
C	12.13
D	15.24

Answer Given By Candidate:**Not Attempted**

Question ID:501165
Topic Name: Chemistry-Section A
Question:
A 42.12% (w/v) solution of NaCl causes precipitation of a certain sol in 10 hours. The coagulating value of NaCl for the sol is
[Given: Molar mass : Na = 23.0 g mol^{-1} ; Cl = 35.5 g mol^{-1}]

A	36 mmol L^{-1}
B	36 mol L^{-1}
C	1440 mol L^{-1}
D	1440 mmol L^{-1}

Answer Given By Candidate:**Not Attempted**

Question ID:501166
Topic Name: Chemistry-Section A

Given below are two statements. One is labelled as **Assertion A** and the other is labelled as **Reason R.**

Assertion A: The first ionization enthalpy for oxygen is lower than that of nitrogen.

Reason R: The four electrons in 2p orbitals of oxygen experience more electron-electron repulsion.

In the light of the above statements, choose the ***correct*** answer from the options given below.

Question:

A	Both **A** and **R** are correct and **R** is the correct explanation of **A.**
B	Both **A** and **R** are correct but **R** is NOT the correct explanation of **A.**
C	**A** is correct but **R** is not correct.
D	**A** is not correct but **R** is correct.

Answer Given By Candidate:**B**

Question ID:501167
Topic Name: Chemistry-Section A

Match **List I** with **List II**.

List I Ore	List II Composition
A. Siderite	I. $FeCO_3$
B. Malachite	II. $CuCO_3.Cu(OH)_2$
C. Sphalerite	III. ZnS
D. Calamine	IV. $ZnCO_3$

Question: Choose the correct answer from the options given below:

A	A-I, B-II, C-III, D-IV
B	A-III, B-IV, C-II, D-I
C	A-IV, B-III, C-I, D-II
D	A-I, B-II, C-IV, D-III

Answer Given By Candidate:**A**

Question ID:501168
Topic Name: Chemistry-Section A

Question: Given below are two statements.

- Statement I: In $CuSO_4 \cdot 5H_2O$, Cu-O bonds are present.
- Statement II: In $CuSO_4 \cdot 5H_2O$, ligands coordinating with Cu(II) ion are O-and S-based ligands.

In the light of the above statements, choose the ***correct*** answer from the options given below.

A	Both Statement I and Statement II are correct.
B	Both Statement I and Statement II are incorrect.
C	Statement I is correct but Statement II is incorrect.
D	Statement I is incorrect but Statement II is correct.

Answer Given By Candidate:**Not Attempted**

Question ID:501169
Topic Name: Chemistry-Section A

Question: Amongst baking soda, caustic soda and washing soda, carbonate anion is present in

A	washing soda only.
B	washing soda and caustic soda only.
C	washing soda and baking soda only.
D	baking soda, caustic soda and washing soda.

Answer Given By Candidate:**C**

Question ID:501170
Topic Name: Chemistry-Section A

Question: Number of lone pair(s) of electrons on central atom and the shape of BrF_3 molecule respectively, are

A	0, triangular planar.
B	1, pyramidal.
C	2, bent T-shape.
D	1, bent T-shape.

Answer Given By Candidate:**C**

Question ID:501171
Topic Name: Chemistry-Section A
Question: Aqueous solution of which of the following boron compounds will be strongly basic in nature?

A $NaBH_4$

B $LiBH_4$

C B_2H_6

D $Na_2B_4O_7$

Answer Given By Candidate:**A**

Question ID:501172
Topic Name: Chemistry-Section A
Question: Sulphur dioxide is one of the components of polluted air. SO_2 is also a major contributor to acid rain. The correct and complete reaction to represent acid rain caused by SO_2 is

A $2\ SO_2 + O_2 \rightarrow 2\ SO_3$

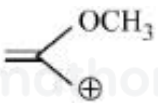
B $SO_2 + O_3 \rightarrow SO_3 + O_2$

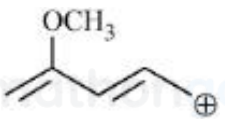
C $SO_2 + H_2O_2 \rightarrow H_2SO_4$

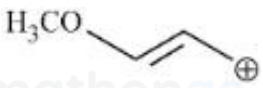
D $2\ SO_2 + O_2 + 2\ H_2O \rightarrow 2H_2SO_4$

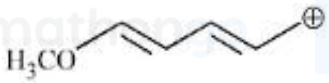
Answer Given By Candidate:**B**

Question ID:501173
Topic Name: Chemistry-Section A
Question: Which of the following carbocations is most stable?

A 

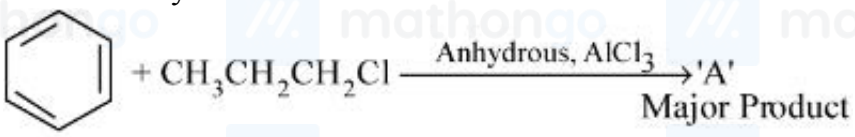
B 

C 

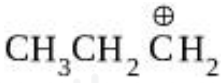
D 

Answer Given By Candidate:**D**

Question ID:501174
Topic Name: Chemistry-Section A

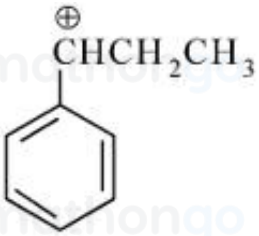


Question: The stable carbocation formed in the above reaction is

A 

B 

C 

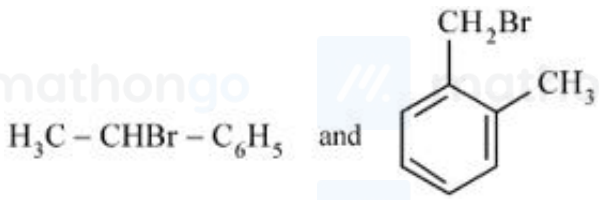
D 

Answer Given By Candidate:**D**

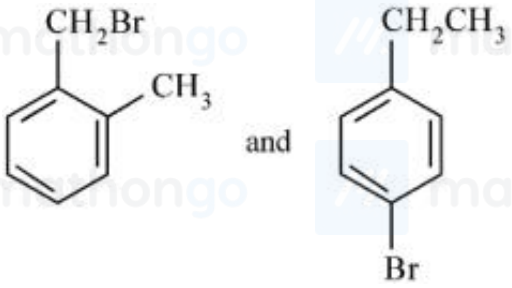
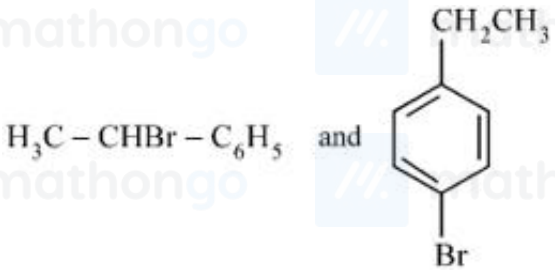
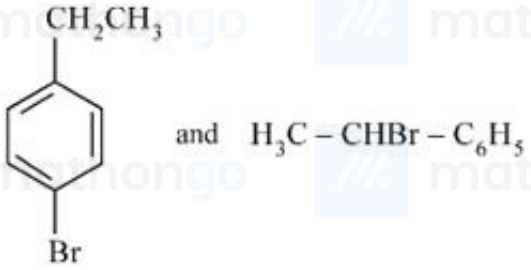
Question ID:501175
Topic Name: Chemistry-Section A

Two isomers (A) and (B) with Molar mass 184 g/mol and elemental composition C, 52.2% ; H, 4.9 % and Br 42.9% gave benzoic acid and p-bromobenzoic acid, respectively on oxidation with $KMnO_4$. Isomer 'A' is optically active and gives a pale yellow precipitate when warmed with alcoholic $AgNO_3$. Isomer 'A' and 'B' are, respectively

Question:

A 

B

	
C	
D	

Answer Given By Candidate:**B**

Question ID:501176
Topic Name: Chemistry-Section A
Question: In Friedel-Crafts alkylation of aniline, one gets

A	alkylated product with ortho and para substitution.
B	secondary amine after acidic treatment.
C	an amide product.
D	positively charged nitrogen at benzene ring.

Answer Given By Candidate:**A**

Question ID:501177
Topic Name: Chemistry-Section A
Question: Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**.
Assertion A: Dacron is an example of polyester polymer.
Reason R: Dacron is made up of ethylene glycol and terephthalic acid monomers.
In the light of the above statements, choose the ***most appropriate*** answer from the options given below.

A	Both **A** and **R** are correct and **R** is the correct explanation of **A.**
B	Both **A** and **R** are correct but **R** is NOT the correct explanation of **A.**
C	**A** is correct but **R** is not correct.
D	**A** is not correct but **R** is correct.

Answer Given By Candidate:**A**

Question ID:501178
Topic Name: Chemistry-Section A
Question: The structure of protein that is unaffected by heating is

A	secondary structure
B	tertiary structure
C	primary structure
D	quaternary structure

Answer Given By Candidate:**C**

Question ID:501179
Topic Name: Chemistry-Section A
Question: The mixture of chloroxylenol and terpineol is an example of

A	antiseptic
B	pesticide
C	disinfectant
D	narcotic analgesic

Answer Given By Candidate:**Not Attempted**

Question ID:501180
Topic Name: Chemistry-Section A
Question: A white precipitate was formed when $BaCl_2$ was added to water extract of an inorganic salt. Further, a gas 'X' with characteristic odour was released when the formed white precipitate was dissolved in dilute HCl. The anion present in the inorganic salt is

A	I^-
B	

	SO_3^{2-}
C	S^{2-}
D	NO_2^-

Answer Given By Candidate:**Not Attempted**

Question ID:501181
Topic Name: Chemistry-Section B

Question: A box contains 0.90 g of liquid water in equilibrium with water vapour at 27°C. The equilibrium vapour pressure of water at 27°C is 32.0 Torr. When the volume of the box is increased, some of the liquid water evaporates to maintain the equilibrium pressure. If all the liquid water evaporates, then the volume of the box must be ____ litre. [nearest integer]
(Given: R = 0.082 L atm K^{-1} mol^{-1})
(Ignore the volume of the liquid water and assume water vapours behave as an ideal gas.)

Answer Given By Candidate:**Not Attempted**

Question ID:501182
Topic Name: Chemistry-Section B

Question: 2.2 g of nitrous oxide (N_2O) gas is cooled at a constant pressure of 1 atm from 310 K to 270 K causing the compression of the gas from 217.1 mL to 167.75 mL. The change in internal energy of the process, ΔU is '−x' J. The value of 'x' is ___. [nearest integer]
(Given: atomic mass of N = 14 g mol^{-1} and of O = 16 g mol^{-1}.
Molar heat capacity of N_2O is 100 J $K^{-1}mol^{-1}$)

Answer Given By Candidate:**Not Attempted**

Question ID:501183
Topic Name: Chemistry-Section B

Question: Elevation in boiling point for 1.5 molal solution of glucose in water is 4 K. The depression in freezing point for 4.5 molal solution of glucose in water is 4 K. The ratio of molal elevation constant to molal depression constant (K_b/K_f) is ___.

Answer Given By Candidate:**3**

Question ID:501184
Topic Name: Chemistry-Section B

Question: The cell potential for the given cell at 298 K
$Pt|H_2(g,1\ bar)|H^+(aq)||Cu^{2+}(aq)|Cu(s)$
is 0.31V. The pH of the acidic solution is found to be 3, whereas the concentration of Cu^{2+} is 10^{-x} M. The value of x is ___.
(Given: $E^{\ominus}_{Cu^{2+}/Cu} = 0.34\ V$ and $\frac{2.303\ RT}{F} = 0.06\ V$)

Answer Given By Candidate:**Not Attempted**

Question ID:501185
Topic Name: Chemistry-Section B
Question: The equation
$k = (6.5\times10^{12}s^{-1})e^{-26000K/T}$
is followed for the decomposition of compound A. The activation energy for the reaction is ___ kJ mol^{-1}. [nearest integer]
(Given: R = 8.314 J $K^{-1}mol^{-1}$)

Answer Given By Candidate:**Not Attempted**

Question ID:501186
Topic Name: Chemistry-Section B
Question: Spin only magnetic moment of $[MnBr_6]^{4-}$ is ___ B.M. (round off to the closest integer)

Answer Given By Candidate:**6**

Question ID:501187
Topic Name: Chemistry-Section B
Question: For the reaction given below :
$CoCl_3 \cdot x\ NH_3 + AgNO_3\ (aq) \rightarrow$
If two equivalents of AgCl precipitate out, then the value of x will be _____.

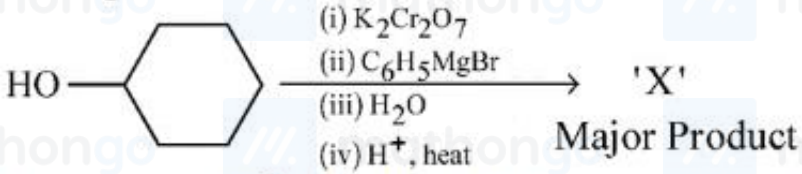
Answer Given By Candidate:**2**

Question ID:501188
Topic Name: Chemistry-Section B
Question: The number of chiral alcohol(s) with molecular formula $C_4H_{10}O$ is _____.

Answer Given By Candidate:**1**

Question ID:501189
Topic Name: Chemistry-Section B

In the given reaction,



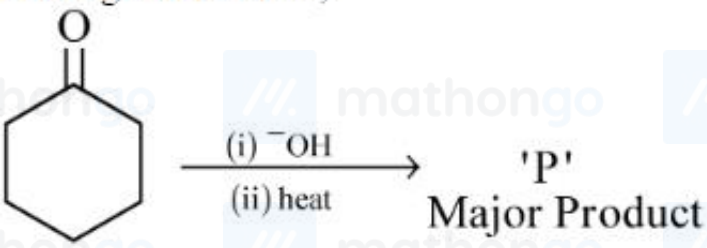
Question: the number of sp^2 hybridised carbon(s) in compound 'X' is ___.

Answer Given By Candidate:**Not Attempted**

Question ID:501190

Topic Name: Chemistry-Section B

In the given reaction,



Question: The number of π electrons present in the product 'P' is ____.

Answer Given By Candidate:**Not Attempted**